

Namespace Dear_ImGui_Sample.Backends

Classes

[AssetLoader](#)

[BedrockImGuiCompatability](#)

[ImGuiImplOpenGL3](#)

Class AssetLoader

Namespace: [Dear ImGui Sample.Backends](#)

Assembly: Osmium-Editor.dll

```
public static class AssetLoader
```

Inheritance

[object](#)  ← AssetLoader

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) ,
[object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) ,
[object.ToString\(\)](#) 

Methods

Font(string, float)

```
public static IntPtr Font(string path, float size)
```

Parameters

path [string](#) 

size [float](#) 

Returns

IntPtr

Image(string)

```
public static Texture Image(string path)
```

Parameters

`path` [string](#) 

Returns

[Texture](#)

Class BedrockImGuiCompatability

Namespace: [Dear_ImGui_Sample.Backends](#)

Assembly: Osmium-Editor.dll

```
public static class BedrockImGuiCompatability
```

Inheritance

[object](#)  ← BedrockImGuiCompatability

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Fields

ScreenSize

```
public static Vector2 ScreenSize
```

Field Value

[Vector2](#) 

Methods

Incorporate()

```
public static void Incorporate()
```

Class ImGuiImplOpenGL3

Namespace: [Dear_ImGui_Sample.Backends](#)

Assembly: Osmium-Editor.dll

```
public static class ImGuiImplOpenGL3
```

Inheritance

[object](#)  ← ImGuiImplOpenGL3

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Methods

Init(bool)

```
public static bool Init(bool useKHRDebugIfAvailable = true)
```

Parameters

useKHRDebugIfAvailable [bool](#) 

Returns

[bool](#) 

NewFrame()

```
public static void NewFrame()
```

RenderDrawData(ImDrawDataPtr)

```
public static void RenderDrawData(ImDrawDataPtr drawData)
```

Parameters

drawData ImDrawDataPtr

SetupRenderState(ImDrawDataPtr, int, int, int)

```
public static void SetupRenderState(ImDrawDataPtr drawData, int fbWidth, int  
fbHeight, int vao)
```

Parameters

drawData ImDrawDataPtr

fbWidth [int](#)

fbHeight [int](#)

vao [int](#)

Shutdown()

```
public static void Shutdown()
```

Namespace Dear_ImGui_Sample.Backends.

Types

Classes

[Texture](#)

Class Texture

Namespace: [Dear ImGui Sample.Backends.Types](#)

Assembly: Osmium-Editor.dll

```
public record Texture : IEquatable<Texture>
```

Inheritance

[object](#)  ← Texture

Implements

[IEquatable](#)  <[Texture](#)>

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

Texture(nint, int, int)

```
public Texture(nint Handle, int Width, int Height)
```

Parameters

Handle [nint](#) 

Width [int](#) 

Height [int](#) 

Properties

Handle

```
public nint Handle { get; init; }
```

Property Value

[nint](#)

Height

```
public int Height { get; init; }
```

Property Value

[int](#)

Width

```
public int Width { get; init; }
```

Property Value

[int](#)

Operators

implicit operator nint(Texture)

```
public static implicit operator nint(Texture texture)
```

Parameters

texture [Texture](#)

Returns

Namespace OsmiumBedrock

Classes

[Bedrock](#)

[Window](#)

Class Bedrock

Namespace: [OsmiumBedrock](#)

Assembly: Osmium-Bedrock.dll

```
public static class Bedrock
```

Inheritance

[object](#)  ← Bedrock

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) ,
[object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) ,
[object.ToString\(\)](#) 

Fields

DeltaTime

```
public static float DeltaTime
```

Field Value

[float](#) 

window

```
public static readonly Window window
```

Field Value

[Window](#)

Methods

Run()

```
public static void Run()
```

Events

Draw

```
public static event Action Draw
```

Event Type

[Action](#)

DrawFinalizer

```
public static event Action DrawFinalizer
```

Event Type

[Action](#)

DrawInitializer

```
public static event Action DrawInitializer
```

Event Type

[Action](#)

Load

`public static event` Action Load

Event Type

[Action](#)

Unload

`public static event` Action Unload

Event Type

[Action](#)

Update

`public static event` Action Update

Event Type

[Action](#)

UpdateFinalizer

`public static event` Action UpdateFinalizer

Event Type

[Action](#)

UpdateInitializer

```
public static event Action UpdateInitializer
```

Event Type

[Action](#)

Class Window

Namespace: [OsmiumBedrock](#)

Assembly: Osmium-Bedrock.dll

```
public sealed class Window : GameWindow, IDisposable
```

Inheritance

[object](#) ← [NativeWindow](#) ← [GameWindow](#) ← Window

Implements

[IDisposable](#)

Inherited Members

[GameWindow.Run\(\)](#), [GameWindow.SwapBuffers\(\)](#), [GameWindow.Close\(\)](#), [GameWindow.TimeSinceLastUpdate\(\)](#), [GameWindow.ResetTimeSinceLastUpdate\(\)](#), [GameWindow.Dispose\(\)](#), [GameWindow.IsMultiThreaded](#), [GameWindow.RenderFrequency](#), [GameWindow.RenderTime](#), [GameWindow.UpdateTime](#), [GameWindow.UpdateFrequency](#), [GameWindow.ExpectedSchedulerPeriod](#), [GameWindow.Load](#), [GameWindow.Unload](#), [GameWindow.UpdateFrame](#), [GameWindow.RenderThreadStarted](#), [GameWindow.RenderFrame](#), [NativeWindow.Focus\(\)](#), [NativeWindow.MakeCurrent\(\)](#), [NativeWindow.ProcessEvents\(double\)](#), [NativeWindow.ProcessWindowEvents\(bool\)](#), [NativeWindow.NewInputFrame\(\)](#), [NativeWindow.PointToClient\(Vector2i\)](#), [NativeWindow.PointToScreen\(Vector2i\)](#), [NativeWindow.IsKeyDown\(Keys\)](#), [NativeWindow.IsKeyPressed\(Keys\)](#), [NativeWindow.IsKeyReleased\(Keys\)](#), [NativeWindow.IsMouseButtonDown\(MouseButton\)](#), [NativeWindow.IsMouseButtonPressed\(MouseButton\)](#), [NativeWindow.IsMouseButtonReleased\(MouseButton\)](#), [NativeWindow.TryGetCurrentMonitorScale\(out float, out float\)](#), [NativeWindow.TryGetCurrentMonitorDpi\(out float, out float\)](#), [NativeWindow.TryGetCurrentMonitorDpiRaw\(out float, out float\)](#), [NativeWindow.CenterWindow\(\)](#), [NativeWindow.CenterWindow\(Vector2i\)](#), [NativeWindow.MakeFullscreen\(MonitorHandle, int?, int?, int?\)](#), [NativeWindow.WindowPtr](#), [NativeWindow.KeyboardState](#), [NativeWindow.JoystickStates](#), [NativeWindow.MousePosition](#), [NativeWindow.MouseState](#), [NativeWindow.IsAnyKeyDown](#), [NativeWindow.IsAnyMouseButtonDown](#), [NativeWindow.VSync](#), [NativeWindow.AutoIconify](#), [NativeWindow.Icon](#), [NativeWindow.IsEventDriven](#), [NativeWindow.ClipboardString](#), [NativeWindow.Title](#), [NativeWindow.API](#), [NativeWindow.Profile](#), [NativeWindow.Flags](#), [NativeWindow.APIVersion](#), [NativeWindow.Context](#), [NativeWindow.CurrentMonitor](#), [NativeWindow.IsFocused](#), [NativeWindow.IsVisible](#), [NativeWindow.Exists](#),

[NativeWindow.IsExiting](#) , [NativeWindow.WindowState](#) , [NativeWindow.WindowBorder](#) ,
[NativeWindow.MousePassthrough](#) , [NativeWindow.AlwaysOnTop](#) , [NativeWindow.Bounds](#) ,
[NativeWindow.Location](#) , [NativeWindow.ClientLocation](#) , [NativeWindow.Size](#) ,
[NativeWindow.ClientSize](#) , [NativeWindow.FramebufferSize](#) , [NativeWindow.MinimumSize](#) ,
[NativeWindow.MaximumSize](#) , [NativeWindow.AspectRatio](#) , [NativeWindow.ClientRectangle](#) ,
[NativeWindow.IsFullscreen](#) , [NativeWindow.Cursor](#) , [NativeWindow.CursorState](#) ,
[NativeWindow.RawMouseInput](#) , [NativeWindow.SupportsRawMouseInput](#) ,
[NativeWindow.HasTransparentFramebuffer](#) , [NativeWindow.Move](#) , [NativeWindow.Resize](#) ,
[NativeWindow.FramebufferResize](#) , [NativeWindow.Refresh](#) , [NativeWindow.Closing](#) ,
[NativeWindow.Minimized](#) , [NativeWindow.Maximized](#) , [NativeWindow.JoystickConnected](#) ,
[NativeWindow.FocusedChanged](#) , [NativeWindow.KeyDown](#) , [NativeWindow.TextInput](#) ,
[NativeWindow.KeyUp](#) , [NativeWindow.MouseLeave](#) , [NativeWindow.MouseEnter](#) ,
[NativeWindow.MouseDown](#) , [NativeWindow.MouseUp](#) , [NativeWindow.MouseMove](#) ,
[NativeWindow.MouseWheel](#) , [NativeWindow.FileDrop](#) , [object.Equals\(object\)](#) ,
[object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#)

Constructors

Window()

```
public Window()
```

Namespace OsmiumEditor

Classes

[Appdata](#)

[Editor](#)

[Project](#)

[ProjectMemory](#)

[ProjectMenu](#)

Class Appdata

Namespace: [OsmiumEditor](#)

Assembly: Osmium-Editor.dll

```
public static class Appdata
```

Inheritance

[object](#)  ← Appdata

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Fields

Path

```
public static readonly string Path
```

Field Value

[string](#) 

Methods

AssureExistence(string)

```
public static void AssureExistence(string __path)
```

Parameters

__path [string](#) 

Class Editor

Namespace: [OsmiumEditor](#)

Assembly: Osmium-Editor.dll

```
public static class Editor
```

Inheritance

[object](#)  ← Editor

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) ,
[object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) ,
[object.ToString\(\)](#) 

Fields

CSProjModuleReferenceEnd

```
public const string CSProjModuleReferenceEnd = "</HintPath>\n</Reference>\n    </ItemGroup>"
```

Field Value

[string](#) 

CSProjModuleReferenceMiddle

```
public const string CSProjModuleReferenceMiddle = "\">\n    <HintPath>"
```

Field Value

[string](#) 

CSProjModuleReferenceStart

```
public const string CSProjModuleReferenceStart = "<ItemGroup>\n<Reference Include=\""
```

Field Value

[string](#)

EditorCompileCSProjEnd

```
public const string EditorCompileCSProjEnd = "\n\n</Project>\n"
```

Field Value

[string](#)

EditorCompileCSProjFront

```
public const string EditorCompileCSProjFront = "<Project\nSdk=\"Microsoft.NET.Sdk\">\n\n  <PropertyGroup>\n<TargetFramework>net10.0</TargetFramework>\n<ImplicitUsings>enable</ImplicitUsings>\n      <Nullable>enable</Nullable>\n<OutputPath>.editorCompileOutputPath</OutputPath>\n  </PropertyGroup> \n\n  <ItemGroup>\n      <PackageReference Include=\"OpenTK\" Version=\"4.9.4\" />\n  </ItemGroup> \n  <PropertyGroup>\n\n</PropertyGroup>"
```

Field Value

[string](#)

EditorCompileCSProjPath

```
public const string EditorCompileCSProjPath = ".editorCompileCSProj"
```

Field Value

[string](#)

EditorCompileOutputPath

```
public const string EditorCompileOutputPath = ".editorCompileOutputPath"
```

Field Value

[string](#)

EditorLoadContext

```
public static AssemblyLoadContext EditorLoadContext
```

Field Value

[AssemblyLoadContext](#)

EditorModules

```
public static readonly List<Assembly> EditorModules
```

Field Value

[List](#) <[Assembly](#)>

GlobalUsage

```
public const string GlobalUsage = "//----- READ ME -----  
-----\n// THIS IS AN AUTO-GENERATED FILE BY OSMIUM! EDITING THIS IS OK,  
BUT\n// BE SURE YOU KNOW WHAT YOU ARE DOING, THE AVERAGE USER HAS NO REASON TO BE  
HERE\n//-----  
\n\n//Automatically references OsmiumNucleus to make life easier!\nglobal using
```

```
OsmiumNucleus;\nglobal using OsmiumBedrock;\n\n//Most IDEs will automatically add  
this file to avoid redundant usage statements for obvious libraries\n//this is added  
to match the user experience of most IDEs\nglobal using System;\nglobal using  
System.Collections.Generic;\nglobal using System.IO;\nglobal using  
System.Linq;\nglobal using System.Net.Http;\nglobal using System.Threading;\nglobal  
using System.Threading.Tasks;\n\n//The word IEnumerator can be confusing for  
beginners! Coroutine is cleaner and more readable\nglobal using Coroutine =  
System.Collections.Generic.IEnumerator<OsmiumNucleus.ICoroutineAction>;"
```

Field Value

[string](#)

IDETrickingCSProjEnd

```
public const string IDETrickingCSProjEnd = "\n\n</Project>\n"
```

Field Value

[string](#)

IDETrickingCSProjFront

```
public const string IDETrickingCSProjFront = "<Project  
Sdk=\"Microsoft.NET.Sdk\">\n\n    <PropertyGroup>\n<TargetFramework>net10.0</TargetFramework>\n<ImplicitUsings>enable</ImplicitUsings>\n                                <Nullable>enable</Nullable>\n</PropertyGroup> \n\n    <ItemGroup>\n                                <PackageReference Include=\"OpenTK\"  
Version=\"4.9.4\" />\n    </ItemGroup> \n\n    \n"
```

Field Value

[string](#)

NetVersion

```
public const string NetVersion = "net10.0"
```

Field Value

[string](#)

RuntimeCompileCSProjEnd

```
public const string RuntimeCompileCSProjEnd = "\n\n</Project>\n"
```

Field Value

[string](#)

RuntimeCompileCSProjFront

```
public const string RuntimeCompileCSProjFront = "<Project  
Sdk=\"Microsoft.NET.Sdk\">\n\n    <PropertyGroup>\n<OutputType>Exe</OutputType>\n        <TargetFramework>net10.0</TargetFramework>\n<ImplicitUsings>enable</ImplicitUsings>\n        <Nullable>enable</Nullable>\n<OutputPath>Build/obj</OutputPath>\n    </PropertyGroup> \n\n    <ItemGroup>\n<PackageReference Include=\"OpenTK\" Version=\"4.9.4\" />\n    </ItemGroup> \n    <PropertyGroup>\n<BaseIntermediateOutputPath>Build/obj</BaseIntermediateOutputPath>\n</PropertyGroup  
>"
```

Field Value

[string](#)

RuntimeCompileCSProjPath

```
public const string RuntimeCompileCSProjPath = ".runtimeCompileCSProj"
```

Field Value

[string](#)

RuntimeCompileOutputPath

```
public const string RuntimeCompileOutputPath = "Build/obj"
```

Field Value

[string](#)

RuntimeModules

```
public static readonly List<Assembly> RuntimeModules
```

Field Value

[List](#) <[Assembly](#)>

TopLevelNugget

```
public const string TopLevelNugget = "using  
System.Reflection;\n\n0smium.Initialize();\n\nforeach (AssemblyName assembly in  
Assembly.GetExecutingAssembly().GetReferencedAssemblies())\n{\n    Assembly.Load(assembly);\n    Console.WriteLine(assembly.FullName);\n}\n\nBedrock.Run(); \n0smium.Run();"
```

Field Value

[string](#)

TopLevelNuggetPath

```
public const string TopLevelNuggetPath = ".topLevelNugget"
```

Field Value

[string](#)

Properties

Assemblies

```
public static List<Assembly> Assemblies { get; }
```

Property Value

[List](#) <[Assembly](#)>

Modules

```
public static List<Assembly> Modules { get; }
```

Property Value

[List](#) <[Assembly](#)>

Methods

CompileProject()

```
public static void CompileProject()
```

Main(string[])

```
public static int Main(string[] __args)
```

Parameters

`__args string[]`

Returns

`int`

RefreshEditor()

```
public static void RefreshEditor()
```

SaveProject()

```
public static void SaveProject()
```

Events

OnSave

```
public static event Action? OnSave
```

Event Type

`Action`

Class Project

Namespace: [OsmiumEditor](#)

Assembly: Osmium-Editor.dll

```
public static class Project
```

Inheritance

[object](#)  ← Project

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Fields

ProjectPath

```
public static string ProjectPath
```

Field Value

[string](#) 

Properties

RuntimeModulesPath

```
public static string RuntimeModulesPath { get; }
```

Property Value

[string](#) 

SourcePath

```
public static string SourcePath { get; }
```

Property Value

[string](#)

Methods

GetProjectSubdirectory(bool, params string[])

```
public static string GetProjectSubdirectory(bool regenerate, params  
string[] subdirectoryPath)
```

Parameters

regenerate [bool](#)

subdirectoryPath [string](#) []

Returns

[string](#)

GetProjectSubpath(bool, params string[])

```
public static string GetProjectSubpath(bool regenerate, params  
string[] subdirectoryPath)
```

Parameters

regenerate [bool](#)

subdirectoryPath [string](#) []

Returns

Class ProjectMemory

Namespace: [OsmiumEditor](#)

Assembly: Osmium-Editor.dll

```
public static class ProjectMemory
```

Inheritance

[object](#)  ← ProjectMemory

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Fields

ProjectRefPath

```
public static string ProjectRefPath
```

Field Value

[string](#) 

Projects

```
public static List<string> Projects
```

Field Value

[List](#)  <[string](#)  >

Methods

AppendProject(string)

```
public static void AppendProject(string __projectPath)
```

Parameters

`__projectPath` [string](#)

ForgetProject(string)

```
public static void ForgetProject(string __projectPath)
```

Parameters

`__projectPath` [string](#)

GetProjectName(string)

```
public static string GetProjectName(string __projectPath)
```

Parameters

`__projectPath` [string](#)

Returns

[string](#)

RefreshProjectList()

```
public static void RefreshProjectList()
```

RefreshProjectTime(string)

```
public static void RefreshProjectTime(string __projectPath)
```

Parameters

`__projectPath` [string](#)

ValidOsmiumProject(string)

```
public static bool ValidOsmiumProject(string __path)
```

Parameters

`__path` [string](#)

Returns

[bool](#)

Class ProjectMenu

Namespace: [OsmiumEditor](#)

Assembly: Osmium-Editor.dll

```
public static class ProjectMenu
```

Inheritance

[object](#)  ← ProjectMenu

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Methods

CreateProjectPrompt()

```
public static void CreateProjectPrompt()
```

Initialize()

```
public static void Initialize()
```

OpenProjectPrompt()

```
public static void OpenProjectPrompt()
```

Namespace OsmiumNucleus

Classes

[AlwaysUpdate](#)

Allows a Component to be updated even if Osmium is only initialized

[Component](#)

The backing class for all Components in Osmium. Inherit this to become one.

[ComponentDocker](#)

Base class for both Scenes and Components. Responsible for Managing hierarchy between Components and Scenes and managing children, has Extensive Component Manipulation Available. Also transfers Time and Carries most of the responsibilities akin to the Component. NOT FOR OUTSIDE INHERITANCE (this is very under the hood, and meant to be mostly hidden), please use docker by inheriting component.

[Debug](#)

[DebugMessage](#)

[Delay](#)

Represents a timed delay in a coroutine, which can be used with yield return.

[EditorOnly](#)

Shows that a piece of code is used only for the Editor, and it should be ignored in the compilation phase

[MarkerAttributes](#)

Provides many handy cosmetic attributes for methods to specify use cases or provide information.

[MarkerAttributes.CalculatedProperty](#)

Shows that the given object is calculated every time it is called! Good to know for performance heavy systems.

[MarkerAttributes.Complexity](#)

Just a way to write the time complexity of a calculated property or method.

[MarkerAttributes.ComplexityON](#)

Just a way to write the time complexity of a calculated property or method.

[MarkerAttributes.Expense](#)

Just a way to write how expensive a calculated property or method can be.

[MarkerAttributes.MethodLambda](#)

Shows that the given method does not actually belong to the object, but instead just calls a lambda to another method.

[MarkerAttributes.UnsafeInternal](#)

Shows that the given object is unsafe (ex. it doesn't check for null values and such, or it doesn't have guardrails based on cases). This is just for internal/private methods to remind myself how to call it :) The reasoning is case by case, but most of the time, it is because all the exposing public methods already check, and double checks would only slow me down

[MarkerAttributes.UnsafePipeline](#)

The given function is missing many guiderails, and it should be used carefully.

[MarkerAttributes.VariableLambda](#)

Shows that the given variable does not actually belong to the object, but instead just calls a lambda to another method.

[OnEditorOpen](#)

Used to mark any method you would like to call once Osmium opens a project. Only static, parameterless, methods may be called. Your code will compile regardless, but your event will be ignored at runtime.

[OnInitialize](#)

Used to mark any method you would like to call once Osmium initializes. Only static, parameterless, methods may be called. Your code will compile regardless, but your event will be ignored at runtime.

[Osmium](#)

Bottom class of Osmium. Carries events from MonoGame into Scenes, and provides OpenTK context.

[Scene](#)

One of the main Osmium entities. Acts like a master folder for components to be stored in. Feel free to inherit this and make a custom scene!

Interfaces

[ICoroutineAction](#)

An action taken during a coroutine.

Enums

[Event](#)

Represents an Event in Osmium.

[MarkerAttributes.Complexity.TimeComplexity](#)

[MarkerAttributes.Expense.ExpenseLevel](#)

[MessageType](#)

Class AlwaysUpdate

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Allows a Component to be updated even if Osmium is only initialized

```
[AttributeUsage(AttributeTargets.Class)]  
public class AlwaysUpdate : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← AlwaysUpdate

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#)  , [Attribute.Match\(object\)](#)  , [Attribute.TypeId](#)  ,

[object.Equals\(object, object\)↗](#) , [object.GetType\(\)↗](#) , [object.MemberwiseClone\(\)↗](#) ,
[object.ReferenceEquals\(object, object\)↗](#) , [object.ToString\(\)↗](#)

Class Component

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

The backing class for all Components in Osmium. Inherit this to become one.

```
public abstract class Component : ComponentDocker, IEnumerable<Component>,
    IEnumerable
```

Inheritance

[object](#)  ← [ComponentDocker](#) ← Component

Implements

[IEnumerable](#)  <[Component](#)>, [IEnumerable](#) 

Inherited Members

[ComponentDocker.components](#), [ComponentDocker.componentPriorityDictionary](#),
[ComponentDocker.componentTypeDictionary](#), [ComponentDocker.componentTagDictionary](#),
[ComponentDocker.this\[int\]](#), [ComponentDocker.GetEnumerator\(\)](#),
[ComponentDocker.ComponentMoved](#), [ComponentDocker.ComponentAdded](#),
[ComponentDocker.ComponentDestroyed](#),
[ComponentDocker.Contains\(params IEnumerable<Component>\)](#),
[ComponentDocker.Contains<TComponent>\(\)](#),
[ComponentDocker.Contains\(params IEnumerable<string>\)](#),
[ComponentDocker.Contains<TComponent>\(params IEnumerable<string>\)](#), [ComponentDocker.Get\(\)](#),
[ComponentDocker.Get<TComponent>\(\)](#), [ComponentDocker.Get\(params IEnumerable<string>\)](#),
[ComponentDocker.Get<TComponent>\(params IEnumerable<string>\)](#), [ComponentDocker.GetAll\(\)](#),
[ComponentDocker.GetAll<TComponent>\(\)](#), [ComponentDocker.GetAll\(params IEnumerable<string>\)](#),
[ComponentDocker.GetAll<TComponent>\(IEnumerable<string>\)](#),
[ComponentDocker.Add\(params IEnumerable<Component>\)](#),
[ComponentDocker.Add<TComponent>\(\)](#), [ComponentDocker.Add<TComponent>\(string\)](#),
[ComponentDocker.Add<TComponent>\(int\)](#), [ComponentDocker.Add<TComponent>\(string,int\)](#),
[ComponentDocker.Destroy\(params IEnumerable<Component>\)](#),
[ComponentDocker.Destroy<TComponent>\(\)](#),
[ComponentDocker.Destroy\(params IEnumerable<string>\)](#),
[ComponentDocker.Destroy<TComponent>\(params IEnumerable<string>\)](#),
[ComponentDocker.DestroyAll\(\)](#), [ComponentDocker.DestroyAll<TComponent>\(\)](#),
[ComponentDocker.DestroyAll\(params IEnumerable<string>\)](#),
[ComponentDocker.DestroyAll<TComponent>\(params IEnumerable<string>\)](#),

[ComponentDock.Move\(ComponentDock, params IEnumerable<Component>\)](#) ,
[ComponentDock.Move<TComponent>\(ComponentDock\)](#) ,
[ComponentDock.Move\(ComponentDock, params IEnumerable<string>\)](#) ,
[ComponentDock.Move<TComponent>\(ComponentDock, params IEnumerable<string>\)](#) ,
[ComponentDock.MoveAll\(ComponentDock\)](#) ,
[ComponentDock.MoveAll<TComponent>\(ComponentDock\)](#) ,
[ComponentDock.MoveAll\(ComponentDock, params IEnumerable<string>\)](#) ,
[ComponentDock.MoveAll<TComponent>\(ComponentDock, params IEnumerable<string>\)](#) ,
[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) ,
[object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#)

Properties

AllParents

All parent Components and the parents of the parents up until the Scene. Will only list parents of parents, not uncle Components. And will not list the Scene

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Medium)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public IReadOnlyList<Component> AllParents { get; }
```

Property Value

[IReadOnlyList](#) <[Component](#)>

ComponentParent

Returns the Parent Component. Will be null if the Component's parent is not a Component.

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.VeryLow)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.01)]  
public Component? ComponentParent { get; }
```

Property Value

[Component](#)

DeltaTime

The seconds since the last type of frame event. For instance if Draw was just called, DeltaTime would currently reflect the time since the last DRAW call, even though there was an Update call in between. Use this for framerate independent logic!

```
protected static double DeltaTime { get; }
```

Property Value

[double](#)

Enabled

If the component receives time events or not.

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.VeryLow)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.01)]  
public bool Enabled { get; set; }
```

Property Value

[bool](#)

Name

The Component's name

```
public string Name { get; set; }
```

Property Value

[string](#)

OsmiumWindow

Bottom class of Osmium. Contains the OpenTK Instance.

```
protected static Window OsmiumWindow { get; }
```

Property Value

[Window](#)

Parent

Current parent of the Component. Can be either a Scene or another Component.

```
public ComponentDock Parent { get; }
```

Property Value

[ComponentDock](#)

Priority

Represents the Component's Update priority; higher priorities get updated first. Can be any integer in the range -63 -> 63. Easy to calculate, but when set, it must resort all Components.

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.High)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public int Priority { get; set; }
```

Property Value

[int](#)

Scene

Scene the Component resides in. To save ram it's not stored but calculated on the fly.

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Medium)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public Scene? Scene { get; }
```

Property Value

[Scene](#)

SceneParent

Returns the Parent Scene. Will be null if the Component's parent is not a Scene.

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.VeryLow)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.01)]  
public Scene? SceneParent { get; }
```

Property Value

[Scene](#)

Scenes

```
protected static IReadOnlySet<Scene> Scenes { get; }
```

Property Value

[IReadOnlySet](#) <[Scene](#)>

Tags

A list of all tags belonging to the component. Use AddTag() to modify it. Easy to calculate, but setting this requires that the Docker resorts Components;

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Medium)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public IReadOnlySet<string> Tags { get; set; }
```

Property Value

[IReadOnlySet](#) <[string](#)>

Methods

AddScene(string)

Creates a new [Scene](#) with the given name, returns null if creating it fails

```
protected static Scene AddScene(string __name)
```

Parameters

__name [string](#)

The name of the new Scene and its unique identifier as a [string](#); cannot match up with any other names in the scene.

Returns

[Scene](#)

The [Scene](#) you created, or null if there is a failure.

AddTag(string)

Adds a new tag to the component

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Low)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public void AddTag(string __tag)
```

Parameters

__tag [string](#)

ContainsScene(string)

Tells you whether a [Scene](#) is currently loaded or not

```
protected static bool ContainsScene(string __name)
```

Parameters

`__name` [string](#)

The name of the [Scene](#) you are searching for

Returns

[bool](#)

A [bool](#) that is either true or false depending on if a scene with the given name is loaded or not

Destroy()

Makes the Component destroy itself

```
public void Destroy()
```

GetScene(string)

Finds a [Scene](#) from the given name.

```
protected static Scene GetScene(string __name)
```

Parameters

`__name` [string](#)

The name of the [Scene](#) you want to find

Returns

[Scene](#)

The newly found [Scene](#)! Or null if there currently isn't one

Move(ComponentDock)

```
public void Move(ComponentDock __newDock)
```

Parameters

__newDock [ComponentDock](#)

RemoveScene(Scene)

Removes a currently loaded [Scene](#) from Osmium

```
protected static void RemoveScene(Scene __scene)
```

Parameters

__scene [Scene](#)

The scene to remove

RemoveScene(string)

```
protected static void RemoveScene(string __name)
```

Parameters

__name [string](#)[↗](#)

RemoveTag(string)

Removes a tag from the component.

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Low)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public void RemoveTag(string __tag)
```

Parameters

`__tag` [string](#)

Start(IEnumerator<ICoroutineAction>)

Begins a coroutine, which will advance asynchronously every frame.

```
public void Start(IEnumerator<ICoroutineAction> __coroutine)
```

Parameters

`__coroutine` [IEnumerator](#) <[ICoroutineAction](#)>

ToString()

Just gives the Component's name. But makes debugging a little easier :).

```
public override string ToString()
```

Returns

[string](#)

Class ComponentDock

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Base class for both Scenes and Components. Responsible for Managing hierarchy between Components and Scenes and managing children, has Extensive Component Manipulation Available. Also transfers Time and Carries most of the responsibilities akin to the Component. NOT FOR OUTSIDE INHERITANCE (this is very under the hood, and meant to be mostly hidden), please use docker by inheriting component.

```
public abstract class ComponentDock : IEnumerable<Component>, IEnumerable
```

Inheritance

[object](#)  ← ComponentDock

Implements

[IEnumerable](#)  <[Component](#)>, [IEnumerable](#) 

Derived

[Component](#), [Scene](#)

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Fields

_componentPriorityDictionary

Holds all the Components sorted by priority. This optimizes priority changes

```
protected readonly SortedDictionary<int, HashSet<Component>>  
_componentPriorityDictionary
```

Field Value

[SortedDictionary](#) [<int](#), [HashSet](#) [<Component](#)>>

_componentTagDictionary

Stores a Component in a list at each of their tags. This optimizes Get(string tag) to O(1)

```
protected readonly Dictionary<string, HashSet<Component>> _componentTagDictionary
```

Field Value

[Dictionary](#) [<string](#), [HashSet](#) [<Component](#)>>

_componentTypeDictionary

Holds a list of Components at each of their types. This optimizes Get<Type> to O(1)

```
protected readonly Dictionary<Type, HashSet<Component>> _componentTypeDictionary
```

Field Value

[Dictionary](#) [<Type](#), [HashSet](#) [<Component](#)>>

_components

Core of the Docker, holds all the Components.

```
protected readonly List<Component> _components
```

Field Value

[List](#) [<Component](#)>

Properties

this[int]

```
public Component this[int __index] { get; set; }
```

Parameters

`__index` [int](#)

Property Value

[Component](#)

Methods

Add(params IEnumerable<Component>)

Adds every [Component](#) in Components as children to the [ComponentDock](#).

```
public void Add(params IEnumerable<Component> Components)
```

Parameters

`Components` [IEnumerable](#) <[Component](#)>

All the [Components](#) to add.

Add<TComponent>()

Adds a new [Component](#) of type TComponent.

```
public TComponent Add<TComponent>() where TComponent : Component, new()
```

Returns

TComponent

The new [Component](#) that was created.

Type Parameters

TComponent

The type of Component to add.

Add<TComponent>(int)

Adds a new [Component](#) of type TComponent with a priority of Priority.

```
public TComponent Add<TComponent>(int Priority) where TComponent : Component, new()
```

Parameters

Priority [int](#)

The priority of the new [Component](#).

Returns

TComponent

The new [Component](#) that was created.

Type Parameters

TComponent

The type of the [Component](#).

Add<TComponent>(string)

Adds a new [Component](#) of type TComponent with the name Name.

```
public TComponent Add<TComponent>(string Name) where TComponent : Component, new()
```

Parameters

Name [string](#)

The name of the [Component](#).

Returns

TComponent

The new [Component](#) that was created.

Type Parameters

TComponent

The type of [Component](#) to add.

Add<TComponent>(string, int)

Adds a new [Component](#) of type TComponent with a priority of Priority and a name of Name.

```
public TComponent Add<TComponent>(string Name, int Priority) where TComponent :  
Component, new()
```

Parameters

Name [string](#)

The name of the [Component](#).

Priority [int](#)

The priority of the new [Component](#)

Returns

TComponent

The new [Component](#) that was created.

Type Parameters

TComponent

The type of [Component](#) to create.

Contains(params IEnumerable<Component>)

Tests to see if every single [Component](#) is a direct child of the [ComponentDock](#).

```
public bool Contains(params IEnumerable<Component> Components)
```

Parameters

Components [IEnumerable](#)<[Component](#)>

The components that are being tested

Returns

[bool](#)

True if every [Component](#) in Components belongs to the Docker, otherwise false.

Contains(params IEnumerable<string>)

Tests to see if the [ComponentDock](#) has a child with all the given tags at the very least.

```
public bool Contains(params IEnumerable<string> Tags)
```

Parameters

Tags [IEnumerable](#)<[string](#)>

The list of tags to find on a [Component](#)

Returns

[bool](#)

True if there is at last one [Component](#) with at least all the given tags, otherwise false.

Contains<TComponent>()

Tests if the [ComponentDock](#) contains a components of type TComponent.

```
public bool Contains<TComponent>() where TComponent : Component, new()
```

Returns

[bool](#)

True if the Docker has at least one child of type TComponent, otherwise false.

Type Parameters

TComponent

The type of the [Component](#) you want to check for

Contains<TComponent>(params IEnumerable<string>)

Tests to see if the [ComponentDocker](#) has a child with at least all the given tags, and of type TComponent.

```
public bool Contains<TComponent>(params IEnumerable<string> Tags) where TComponent :  
Component, new()
```

Parameters

Tags [IEnumerable](#) <[string](#)>

The list of tags to find on a Component

Returns

[bool](#)

True if there is at least one [Component](#) of type TComponent and with all tags in Tags

Type Parameters

TComponent

The type of [Component](#) to search for

Destroy(params IEnumerable<Component>)

Destroys all [Components](#) in Components.

```
public void Destroy(params IEnumerable<Component> Components)
```

Parameters

Components [IEnumerable](#) <[Component](#)>

The list of [Components](#) to destroy.

Destroy(params IEnumerable<string>)

Destroys the first found [Components](#) that has all the given tags.

```
public void Destroy(params IEnumerable<string> Tags)
```

Parameters

Tags [IEnumerable](#) <[string](#)>

The tags to search for.

DestroyAll()

Destroys all [Components](#) owned by the [ComponentDock](#).

```
public void DestroyAll()
```

DestroyAll(params IEnumerable<string>)

Destroys all [Components](#) with all the given tags.

```
public void DestroyAll(params IEnumerable<string> Tags)
```

Parameters

Tags [IEnumerable](#) <[string](#)>

The tags to search for.

DestroyAll<TComponent>()

Destroys all [Components](#) of type TComponent.

```
public void DestroyAll<TComponent>() where TComponent : Component, new()
```

Type Parameters

TComponent

The type of [Component](#) to destroy.

DestroyAll<TComponent>(params IEnumerable<string>)

Destroys all [Components](#) of type TComponent with the given tags.

```
public void DestroyAll<TComponent>(params IEnumerable<string> Tags) where TComponent : Component, new()
```

Parameters

Tags [IEnumerable](#) <[string](#)>

The tags to search for.

Type Parameters

TComponent

The type of [Component](#) to destroy.

Destroy<TComponent>()

Destroys the first found [Components](#) of type TComponent.

```
public void Destroy<TComponent>() where TComponent : Component, new()
```

Type Parameters

TComponent

The type of [Component](#) to destroy.

Destroy<TComponent>(params IEnumerable<string>)

Destroys the first found [Component](#) of type TComponent with all the given tags.

```
public void Destroy<TComponent>(params IEnumerable<string> Tags) where TComponent : Component, new()
```

Parameters

Tags [IEnumerable](#)<[string](#)>

The tags to search for.

Type Parameters

TComponent

The type of component to search for.

Get()

Finds the very first [Component](#) owned by the [ComponentDocker](#) that belongs to the [Component Docker](#).

```
public Component? Get()
```

Returns

[Component](#)

The very first [Component](#) owned by the [ComponentDock](#) found in the [ComponentDock](#), null if there is none.

Get(params IEnumerable<string>)

Finds the very first [Component](#) owned by the [ComponentDock](#) that has all the tags in Tags.

```
public Component? Get(params IEnumerable<string> Tags)
```

Parameters

Tags [IEnumerable](#) <[string](#)>

The tags to search for

Returns

[Component](#)

The first [Component](#) owned by the [ComponentDock](#) that has all the tags present in Tags, null if there is none.

GetAll()

Finds all [Components](#) that belong to the [ComponentDock](#).

```
public List<Component> GetAll()
```

Returns

[List](#) <[Component](#)>

Returns a [List<T>](#) of all [Components](#) that belongs to the [ComponentDock](#).

GetAll(params IEnumerable<string>)

Finds all [Components](#) owned by the [ComponentDocker](#) that have all the tags in Tags.

```
public List<Component> GetAll(params IEnumerable<string> Tags)
```

Parameters

Tags [IEnumerable](#)<[string](#)>

All the tags to search for.

Returns

[List](#)<[Component](#)>

A [List](#)<[T](#)> of all [Components](#) that belong to the [ComponentDocker](#) and have all the tags in Tags.

GetAll<TComponent>()

Finds all [Components](#) owned by the [ComponentDocker](#) of type TComponent.

```
public List<TComponent> GetAll<TComponent>() where TComponent : Component, new()
```

Returns

[List](#)<[TComponent](#)>

A [List](#)<[T](#)> of all [Components](#) that belong to the [ComponentDocker](#) and are of type TComponent.

Type Parameters

TComponent

The type of [Component](#) to search for.

GetAll<TComponent>(IEnumerable<string>)

Finds all the [Components](#) owned by the [ComponentDocker](#) of type TComponent that have all the tags in Tags.

```
public List<TComponent> GetAll<TComponent>(IEnumerable<string> Tags) where  
TComponent : Component, new()
```

Parameters

Tags [IEnumerable](#)<[string](#)>

All the tags to search for.

Returns

[List](#)<TComponent>

A [List](#)<T> of all [Components](#) that belong to the [ComponentDocker](#) of type TComponent and have all the tags in Tags.

Type Parameters

TComponent

The type of [Component](#) to search for .

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Component> GetEnumerator()
```

Returns

[IEnumerator](#)<[Component](#)>

An enumerator that can be used to iterate through the collection.

Get<TComponent>()

Finds the very first [Component](#) owned by the [ComponentDocker](#) that is of type TComponent.

```
public TComponent? Get<TComponent>() where TComponent : Component, new()
```

Returns

TComponent

The very first [Component](#) owned by the [ComponentDocker](#) of type TComponent, null if there is none.

Type Parameters

TComponent

The true of [Component](#) to search for

Get<TComponent>(params IEnumerable<string>)

Finds the first [Component](#) owned by the [ComponentDocker](#) that is of type TComponent and has all the tags in Tags.

```
public TComponent? Get<TComponent>(params IEnumerable<string> Tags) where TComponent : Component, new()
```

Parameters

Tags [IEnumerable](#)<[string](#)>

The tags to search for

Returns

TComponent

The first [Component](#) owned by the [ComponentDocker](#) of type TComponent and has all the tags present in Tags, null if there is none.

Type Parameters

TComponent

The type of [Component](#) to search for

Move(ComponentDock, params IEnumerable<Component>)

Moves all the [Components](#) in Components to the given [ComponentDock](#).

```
public void Move(ComponentDock ComponentDock, params IEnumerable<Component>
Components)
```

Parameters

ComponentDock [ComponentDock](#)

The dock to move the [Components](#) to.

Components [IEnumerable](#) <[Component](#)>

the [Components](#) to move to the other [ComponentDock](#)

Move(ComponentDock, params IEnumerable<string>)

Moves the first found [Component](#) with all the given tags, to the given [ComponentDock](#).

```
public void Move(ComponentDock ComponentDock, params IEnumerable<string> __tags)
```

Parameters

ComponentDock [ComponentDock](#)

The [ComponentDock](#) to move the [Component](#) to.

__tags [IEnumerable](#) <[string](#)>

The tags to search for.

MoveAll(ComponentDock)

Moves all [Components](#) to the given [ComponentDock](#).

```
public void MoveAll(ComponentDock ComponentDock)
```

Parameters

ComponentDock [ComponentDock](#)

The [ComponentDock](#) to move all [Components](#) to.

MoveAll(ComponentDock, params IEnumerable<string>)

Moves all [Components](#) with the given tags, to the [ComponentDock](#).

```
public void MoveAll(ComponentDock ComponentDock, params  
IEnumerable<string> Tags)
```

Parameters

ComponentDock [ComponentDock](#)

The [ComponentDock](#) to move the [Components](#) to.

Tags [IEnumerable](#) <[string](#)>

The tags to search for.

MoveAll<TComponent>(ComponentDock)

Moves all [Components](#) of type TComponent to the given [ComponentDock](#).

```
public void MoveAll<TComponent>(ComponentDock ComponentDock) where TComponent :  
Component, new()
```

Parameters

ComponentDock [ComponentDock](#)

The [ComponentDock](#) to move the [Components](#) to.

Type Parameters

TComponent

The type of [Components](#) to move.

MoveAll<TComponent>(ComponentDock, params IEnumerable<string>)

Moves all [Components](#) of type TComponent to the [ComponentDock](#).

```
public void MoveAll<TComponent>(ComponentDock ComponentDock, params  
IEnumerable<string> Tags) where TComponent : Component, new()
```

Parameters

ComponentDock [ComponentDock](#)

The [ComponentDock](#) to move the [Components](#) to.

Tags [IEnumerable](#)<[string](#)>

The tags to search for.

Type Parameters

TComponent

The type of [Component](#) to search for.

Move<TComponent>(ComponentDock)

Moves the first [Component](#) of type TComponent.

```
public void Move<TComponent>(ComponentDock ComponentDock) where TComponent :  
Component, new()
```

Parameters

ComponentDocker [ComponentDocker](#)

The [ComponentDocker](#) to move the [Component](#) to.

Type Parameters

TComponent

The type of [Component](#) to search for.

Move<TComponent>(ComponentDocker, params IEnumerable<string>)

Moves the first found [Component](#) of type TComponent to with all the given tags, to the given [ComponentDocker](#).

```
public void Move<TComponent>(ComponentDocker ComponentDocker, params  
IEnumerable<string> __tags) where TComponent : Component, new()
```

Parameters

ComponentDocker [ComponentDocker](#)

The [ComponentDocker](#) to move to the [Component](#) to.

__tags [IEnumerable](#) <[string](#)>

The tags to search for.

Type Parameters

TComponent

The type of [Component](#).

Events

ComponentAdded

```
public static event Action<ComponentDock, Component>? ComponentAdded
```

Event Type

[Action](#) <[ComponentDock](#), [Component](#)>

ComponentDestroyed

```
public static event Action<ComponentDock, Component>? ComponentDestroyed
```

Event Type

[Action](#) <[ComponentDock](#), [Component](#)>

ComponentMoved

```
public static event Action<ComponentDock, ComponentDock,  
Component>? ComponentMoved
```

Event Type

[Action](#) <[ComponentDock](#), [ComponentDock](#), [Component](#)>

Class Debug

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

```
public static class Debug
```

Inheritance

[object](#)  ← Debug

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) ,
[object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) ,
[object.ToString\(\)](#) 

Fields

DefaultCategory

```
public const string DefaultCategory = "GENERAL"
```

Field Value

[string](#) 

Properties

ThrowErrors

```
public static bool ThrowErrors { get; set; }
```

Property Value

[bool](#) 

WriteToConsole

```
public static bool WriteToConsole { get; set; }
```

Property Value

[bool](#)

Methods

Clear()

```
public static void Clear()
```

CollectVirtualization()

```
public static void CollectVirtualization()
```

Error(object)

```
public static void Error(object Message)
```

Parameters

Message [object](#)

Error(object, string[], object[])

```
public static void Error(object Message, string[] DetailInfo, object[] Details)
```

Parameters

Message [object](#)

DetailInfo [string](#) []

Details [object](#) []

Fatal(object)

```
public static void Fatal(object Message)
```

Parameters

Message [object](#)

Fatal(object, string[], object[])

```
public static void Fatal(object Message, string[] DetailInfo, object[] Details)
```

Parameters

Message [object](#)

DetailInfo [string](#) []

Details [object](#) []

FindCategory(Assembly)

```
public static string FindCategory(Assembly __callingAssembly)
```

Parameters

__callingAssembly [Assembly](#)

Returns

[string](#)

Log(object)

```
public static void Log(object Message)
```

Parameters

Message [object](#)

Log(object, string[], object[])

```
public static void Log(object Message, string[] DetailInfo, object[] Details)
```

Parameters

Message [object](#)

DetailInfo [string](#) []

Details [object](#) []

Message(DebugMessage)

```
public static void Message(DebugMessage message)
```

Parameters

message [DebugMessage](#)

RegisterCategory(string)

```
public static void RegisterCategory(string __category)
```

Parameters

`__category` [string](#)

Warning(object)

```
public static void Warning(object Message)
```

Parameters

`Message` [object](#)

Warning(object, string[], object[])

```
public static void Warning(object Message, string[] DetailInfo, object[] Details)
```

Parameters

`Message` [object](#)

`DetailInfo` [string](#) []

`Details` [object](#) []

Events

OnClear

```
public static event Action? OnClear
```

Event Type

[Action](#)

OnMessage

```
public static event Action<DebugMessage>? OnMessage
```

Event Type

[Action](#)  [<DebugMessage>](#)

Class DebugMessage

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

```
public class DebugMessage
```

Inheritance

[object](#)  ← DebugMessage

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) ,
[object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) 

Constructors

DebugMessage(string, string, string[], object[], MessageType)

```
public DebugMessage(string category, string message, string[] detailInfo, object[]  
details, MessageType __type)
```

Parameters

category [string](#) 

message [string](#) 

detailInfo [string](#)  []

details [object](#)  []

__type [MessageType](#)

Properties

CallStack

```
public virtual StackTrace CallStack { get; }
```

Property Value

[StackTrace](#)

Category

```
public virtual string Category { get; }
```

Property Value

[string](#)

DetailInfo

```
public virtual string[] DetailInfo { get; }
```

Property Value

[string](#) []

Details

```
public virtual object[] Details { get; }
```

Property Value

[object](#) []

IsError

```
public virtual bool IsError { get; }
```

Property Value

[bool](#)

IsFatal

```
public virtual bool IsFatal { get; }
```

Property Value

[bool](#)

IsInfo

```
public virtual bool IsInfo { get; }
```

Property Value

[bool](#)

IsWarning

```
public virtual bool IsWarning { get; }
```

Property Value

[bool](#)

Message

```
public virtual string Message { get; }
```

Property Value

[string](#) 

MessageType

```
public virtual MessageType MessageType { get; }
```

Property Value

[MessageType](#)

Time

```
public virtual DateTime Time { get; }
```

Property Value

[DateTime](#) 

Methods

ToFullString()

```
public virtual string ToFullString()
```

Returns

[string](#) 

ToSimpleString()

```
public virtual string ToSimpleString()
```

Returns

[string](#) 

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

[string](#) 

A string that represents the current object.

Class Delay

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Represents a timed delay in a coroutine, which can be used with yield return.

```
public class Delay : ICoroutineAction
```

Inheritance

[object](#)  ← Delay

Implements

[ICoroutineAction](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

Delay(TimeSpan)

Represents a timed delay in a coroutine, which can be used with yield return.

```
public Delay(TimeSpan duration)
```

Parameters

duration [TimeSpan](#) 

The amount of time before the delay ends

Methods

Continue()

Processes the current Action in the coroutine

```
public bool Continue()
```

Returns

[bool](#)

Returns false if the coroutine should exit from this action.

FromDays(float)

Creates a new delay from the given time in days

```
public static Delay FromDays(float __days)
```

Parameters

__days [float](#)

Returns

[Delay](#)

FromHours(float)

Creates a new delay from the given time in hours

```
public static Delay FromHours(float __hours)
```

Parameters

__hours [float](#)

Returns

[Delay](#)

FromMilliseconds(float)

Creates a new delay from the given time in milliseconds

```
public static Delay FromMilliseconds(float __milliseconds)
```

Parameters

`__milliseconds` [float](#)

Returns

[Delay](#).

FromMinutes(float)

Creates a new delay from the given time in minutes

```
public static Delay FromMinutes(float __minutes)
```

Parameters

`__minutes` [float](#)

Returns

[Delay](#).

FromSeconds(float)

Creates a new delay from the given time in seconds

```
public static Delay FromSeconds(float __seconds)
```

Parameters

`__seconds` [float](#)

Returns

[Delay](#)

Class EditorOnly

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Editor.dll

Shows that a piece of code is used only for the Editor, and it should be ignored in the compilation phase

```
[AttributeUsage(AttributeTargets.Class|AttributeTargets.Struct|AttributeTargets.Enum|AttributeTargets.Constructor|AttributeTargets.Method|AttributeTargets.Property|AttributeTargets.Field|AttributeTargets.Event|AttributeTargets.Interface|AttributeTargets.Delegate)]  
public class EditorOnly : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← EditorOnly

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,

[Attribute.IsDefined\(Module, Type, bool\)](#) , [Attribute.IsDefined\(ParameterInfo, Type\)](#) ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#) , [Attribute.Match\(object\)](#) , [Attribute.TypeId](#) ,
[object.Equals\(object, object\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#)

Enum Event

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Represents an Event in Osmium.

```
public enum Event
```

Fields

Create = 4

Destroy = 5

Draw = 3

Load = 0

Unload = 1

Update = 2

Interface ICoroutineAction

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

An action taken during a coroutine.

```
public interface ICoroutineAction
```

Methods

Continue()

Processes the current Action in the coroutine

```
bool Continue()
```

Returns

[bool](#)

Returns false if the coroutine should exit from this action.

Class MarkerAttributes

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Provides many handy cosmetic attributes for methods to specify use cases or provide information.

```
public static class MarkerAttributes
```

Inheritance

[object](#)  ← MarkerAttributes

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Class MarkerAttributes.CalculatedProperty

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Shows that the given object is calculated every time it is called! Good to know for performance heavy systems.

```
[AttributeUsage(AttributeTargets.All)]  
public class MarkerAttributes.CalculatedProperty : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← MarkerAttributes.CalculatedProperty

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#)  , [Attribute.Match\(object\)](#)  , [Attribute.TypeId](#)  ,

[object.Equals\(object, object\).↗](#) , [object.GetType\(\).↗](#) , [object.MemberwiseClone\(\).↗](#) ,
[object.ReferenceEquals\(object, object\).↗](#) , [object.ToString\(\).↗](#)

Class MarkerAttributes.Complexity

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Just a way to write the time complexity of a calculated property or method.

```
[AttributeUsage(AttributeTargets.All)]  
public class MarkerAttributes.Complexity : Attribute
```

Inheritance

[object](#) ← [Attribute](#) ← MarkerAttributes.Complexity

Inherited Members

[Attribute.Equals\(object\)](#) , [Attribute.GetCustomAttribute\(Assembly, Type\)](#) ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#) ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#) ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#) ,
[Attribute.GetCustomAttribute\(Module, Type\)](#) , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#) ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#) ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#) ,
[Attribute.GetCustomAttributes\(Assembly\)](#) , [Attribute.GetCustomAttributes\(Assembly, bool\)](#) ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#) ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#) ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#) , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#) ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#) ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#) , [Attribute.GetCustomAttributes\(Module\)](#) ,
[Attribute.GetCustomAttributes\(Module, bool\)](#) , [Attribute.GetCustomAttributes\(Module, Type\)](#) ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#) ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#) ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#) ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#) ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#) , [Attribute.GetHashCode\(\)](#) ,
[Attribute.IsDefaultAttribute\(\)](#) , [Attribute.IsDefined\(Assembly, Type\)](#) ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#) , [Attribute.IsDefined\(MemberInfo, Type\)](#) ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#) , [Attribute.IsDefined\(Module, Type\)](#) ,
[Attribute.IsDefined\(Module, Type, bool\)](#) , [Attribute.IsDefined\(ParameterInfo, Type\)](#) ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#) , [Attribute.Match\(object\)](#) , [Attribute.TypeId](#) ,

[object.Equals\(object, object\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

Complexity(TimeComplexity)

Just a way to write the time complexity of a calculated property or method.

```
public Complexity(MarkerAttributes.Complexity.TimeComplexity __complexity)
```

Parameters

__complexity [MarkerAttributes.Complexity.TimeComplexity](#)

Fields

complexity

```
public MarkerAttributes.Complexity.TimeComplexity complexity
```

Field Value

[MarkerAttributes.Complexity.TimeComplexity](#)

Enum

MarkerAttributes.Complexity.TimeComplexity

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

```
public enum MarkerAttributes.Complexity.TimeComplexity
```

Fields

`O1 = 0`

`OLogN = 1`

`ON = 2`

`ON2 = 4`

`ON3 = 5`

`ONK = 6`

`ONLogN = 3`

Class MarkerAttributes.ComplexityON

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Just a way to write the time complexity of a calculated property or method.

```
[AttributeUsage(AttributeTargets.All)]  
public class MarkerAttributes.ComplexityON : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← MarkerAttributes.ComplexityON

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#)  , [Attribute.Match\(object\)](#)  , [Attribute.TypeId](#)  ,

[object.Equals\(object, object\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#),
[object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

ComplexityON()

Just a way to write the time complexity of a calculated property or method.

```
public ComplexityON()
```

Class MarkerAttributes.Expense

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Just a way to write how expensive a calculated property or method can be.

```
[AttributeUsage(AttributeTargets.All)]  
public class MarkerAttributes.Expense : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← MarkerAttributes.Expense

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#)  , [Attribute.Match\(object\)](#)  , [Attribute.TypeId](#)  ,

[object.Equals\(object, object\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

Expense(ExpenseLevel)

Just a way to write how expensive a calculated property or method can be.

```
public Expense(MarkerAttributes.Expense.ExpenseLevel __expense)
```

Parameters

`__expense` [MarkerAttributes.Expense.ExpenseLevel](#)

Fields

expense

```
public MarkerAttributes.Expense.ExpenseLevel expense
```

Field Value

[MarkerAttributes.Expense.ExpenseLevel](#)

Enum MarkerAttributes.Expense.ExpenseLevel

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

```
public enum MarkerAttributes.Expense.ExpenseLevel
```

Fields

High = 4

Low = 2

Medium = 3

None = 0

VeryHigh = 5

VeryLow = 1

Class MarkerAttributes.MethodLambda

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Shows that the given method does not actually belong to the object, but instead just calls a lambda to another method.

```
[AttributeUsage(AttributeTargets.All)]  
public class MarkerAttributes.MethodLambda : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← MarkerAttributes.MethodLambda

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#)  , [Attribute.Match\(object\)](#)  , [Attribute.TypeId](#)  ,

[object.Equals\(object, object\)↗](#) , [object.GetType\(\)↗](#) , [object.MemberwiseClone\(\)↗](#) ,
[object.ReferenceEquals\(object, object\)↗](#) , [object.ToString\(\)↗](#)

Class MarkerAttributes.UnsafeInternal

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Shows that the given object is unsafe (ex. it doesn't check for null values and such, or it doesn't have guardrails based on cases). This is just for internal/private methods to remind myself how to call it :) The reasoning is case by case, but most of the time, it is because all the exposing public methods already check, and double checks would only slow me down

```
[AttributeUsage(AttributeTargets.All)]  
public class MarkerAttributes.UnsafeInternal : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← MarkerAttributes.UnsafeInternal

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,

[Attribute.IsDefined\(Module, Type, bool\)](#) , [Attribute.IsDefined\(ParameterInfo, Type\)](#) ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#) , [Attribute.Match\(object\)](#) , [Attribute.TypeId](#) ,
[object.Equals\(object, object\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#)

Class MarkerAttributes.UnsafePipeline

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

The given function is missing many guiderails, and it should be used carefully.

```
[AttributeUsage(AttributeTargets.All)]  
public class MarkerAttributes.UnsafePipeline : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← MarkerAttributes.UnsafePipeline

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#)  , [Attribute.Match\(object\)](#)  , [Attribute.TypeId](#)  ,

[object.Equals\(object, object\)↗](#) , [object.GetType\(\)↗](#) , [object.MemberwiseClone\(\)↗](#) ,
[object.ReferenceEquals\(object, object\)↗](#) , [object.ToString\(\)↗](#)

Class MarkerAttributes.VariableLambda

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Shows that the given variable does not actually belong to the object, but instead just calls a lambda to another method.

```
[AttributeUsage(AttributeTargets.All)]  
public class MarkerAttributes.VariableLambda : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← MarkerAttributes.VariableLambda

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#)  , [Attribute.Match\(object\)](#)  , [Attribute.TypeId](#)  ,

[object.Equals\(object, object\).↗](#) , [object.GetType\(\).↗](#) , [object.MemberwiseClone\(\).↗](#) ,
[object.ReferenceEquals\(object, object\).↗](#) , [object.ToString\(\).↗](#)

Enum MessageType

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

[Flags]

```
public enum MessageType : byte
```

Fields

Error = 4

Fatal = 8

Info = 1

Warning = 2

Class OnEditorOpen

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Editor.dll

Used to mark any method you would like to call once Osmium opens a project. Only static, parameterless, methods may be called. Your code will compile regardless, but your event will be ignored at runtime.

```
[AttributeUsage(AttributeTargets.Method)]  
public class OnEditorOpen : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← OnEditorOpen

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,

[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#) , [Attribute.Match\(object\)](#) , [Attribute.TypeId](#) ,
[object.Equals\(object, object\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#)

Class OnInitialize

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Used to mark any method you would like to call once Osmium initializes. Only static, parameterless, methods may be called. Your code will compile regardless, but your event will be ignored at runtime.

```
[AttributeUsage(AttributeTargets.Method)]  
public class OnInitialize : Attribute
```

Inheritance

[object](#)  ← [Attribute](#)  ← OnInitialize

Inherited Members

[Attribute.Equals\(object\)](#)  , [Attribute.GetCustomAttribute\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttribute\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(MemberInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(Module, Type\)](#)  , [Attribute.GetCustomAttribute\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttribute\(ParameterInfo, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly\)](#)  , [Attribute.GetCustomAttributes\(Assembly, bool\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Assembly, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo\)](#)  , [Attribute.GetCustomAttributes\(MemberInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(MemberInfo, Type, bool\)](#)  , [Attribute.GetCustomAttributes\(Module\)](#)  ,
[Attribute.GetCustomAttributes\(Module, bool\)](#)  , [Attribute.GetCustomAttributes\(Module, Type\)](#)  ,
[Attribute.GetCustomAttributes\(Module, Type, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, bool\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type\)](#)  ,
[Attribute.GetCustomAttributes\(ParameterInfo, Type, bool\)](#)  , [Attribute.GetHashCode\(\)](#)  ,
[Attribute.IsDefaultAttribute\(\)](#)  , [Attribute.IsDefined\(Assembly, Type\)](#)  ,
[Attribute.IsDefined\(Assembly, Type, bool\)](#)  , [Attribute.IsDefined\(MemberInfo, Type\)](#)  ,
[Attribute.IsDefined\(MemberInfo, Type, bool\)](#)  , [Attribute.IsDefined\(Module, Type\)](#)  ,
[Attribute.IsDefined\(Module, Type, bool\)](#)  , [Attribute.IsDefined\(ParameterInfo, Type\)](#)  ,
[Attribute.IsDefined\(ParameterInfo, Type, bool\)](#)  , [Attribute.Match\(object\)](#)  , [Attribute.TypeId](#)  ,

[object.Equals\(object, object\).↗](#) , [object.GetType\(\).↗](#) , [object.MemberwiseClone\(\).↗](#) ,
[object.ReferenceEquals\(object, object\).↗](#) , [object.ToString\(\).↗](#)

Class Osmium

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

Bottom class of Osmium. Carries events from MonoGame into Scenes, and provides OpenTK context.

```
public static class Osmium
```

Inheritance

[object](#)  ← Osmium

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Fields

SafeEscape

If Osmium crashes, with SafeEscape enabled, it will still allow exiting statements to be run

```
public static bool SafeEscape
```

Field Value

[bool](#) 

Version

Current Version of Osmium

```
public const string Version = "1.3"
```

Field Value

[string](#)

Properties

Assemblies

All assemblies loaded by Osmium

```
public static IReadOnlyList<Assembly> Assemblies { get; }
```

Property Value

[IReadOnlyList](#) <[Assembly](#)>

DeltaTime

The seconds since the last type of frame event. For instance if Draw was just called, DeltaTime would currently reflect the time since the last DRAW call, even though there was an Update call in between. Use this for framerate independent logic!

```
public static float DeltaTime { get; }
```

Property Value

[float](#)

IsClosed

Displays if Osmium has closed or not.

```
public static bool IsClosed { get; }
```

Property Value

[bool](#)

IsInitialized

Displays if Osmium has Started or not.

```
public static bool IsInitialized { get; }
```

Property Value

[bool](#)

IsRunning

Displays if the update loop is active.

```
public static bool IsRunning { get; }
```

Property Value

[bool](#)

IsVirtualized

Displays if Osmium is running virtually

```
public static bool IsVirtualized { get; }
```

Property Value

[bool](#)

Scenes

List of all scenes currently loaded in the kernel.

```
public static IReadOnlySet<Scene> Scenes { get; }
```

Property Value

[IReadOnlySet](#) <[Scene](#)>

Window

Bottom class of Osmium. Contains the OpenTK Instance.

```
public static Window Window { get; }
```

Property Value

[Window](#)

Methods

AddScene(Scene)

Adds a new [Scene](#) to Osmium!

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Low)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.01)]  
public static void AddScene(Scene __scene)
```

Parameters

__scene [Scene](#)

The [Scene](#) you want to add

Remarks

Using this is only recommended for custom types of Scene! If you wish to Spawn in a normal scene, use [AddScene\(string\)](#).

AddScene(string)

Creates a new [Scene](#) with the given name, returns null if creating it fails

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Low)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.01)]  
public static Scene? AddScene(string __name)
```

Parameters

__name [string](#)

The name of the new Scene and its unique identifier as a [string](#); cannot match up with any other names in the scene.

Returns

[Scene](#)

The [Scene](#) you created, or null if there is a failure.

Close()

Closes Osmium and ends the OpenTK instance!

```
public static void Close()
```

ContainsScene(Scene)

Tells you whether a [Scene](#) is currently loaded or not

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Medium)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.01)]  
public static bool ContainsScene(Scene __scene)
```

Parameters

__scene [Scene](#)

A reference to the [Scene](#) you are looking for

Returns

[bool](#)

A [bool](#) that is either true or false depending on if a scene with the given name is loaded or not

ContainsScene(string)

Tells you whether a [Scene](#) is currently loaded or not

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Medium)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public static bool ContainsScene(string __name)
```

Parameters

__name [string](#)

The name of the [Scene](#) you are searching for

Returns

[bool](#)

A [bool](#) that is either true or false depending on if a scene with the given name is loaded or not

EditorInitialize()

Initializes the Context and marks Osmium as initialized; but does not Resolve types

```
public static void EditorInitialize()
```

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for Radium! So don't use it unless you know what you are doing. These methods are made required in order to use Virtualization! Use Editor Methods instead of normal ones for Virtualization to work.

EditorRun()

Starts OpenTK but doesn't let the update loop run!

```
public static void EditorRun()
```

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for Radium! So don't use it unless you know what you are doing. These methods are made required in order to use Virtualization! Use Editor Methods instead of normal ones for Virtualization to work.

GetScene(string)

Finds a [Scene](#) from the given name.

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Medium)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public static Scene? GetScene(string __name)
```

Parameters

__name [string](#)

The name of the [Scene](#) you want to find

Returns

[Scene](#)

The newly found [Scene](#)! Or null if there currently isn't one

Initialize()

Starts up Osmium, and resolves all the Components in the App Domain! Must be called before Run() and Osmium is not guaranteed to work until you call this method!

```
public static void Initialize()
```

RemoveScene(Scene)

Removes a currently loaded [Scene](#) from Osmium

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Medium)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.01)]  
public static void RemoveScene(Scene __scene)
```

Parameters

__scene [Scene](#)

The scene to remove

RemoveScene(string)

Removes a currently loaded [Scene](#) from Osmium from a given name

```
[MarkerAttributes.Expense(MarkerAttributes.Expense.ExpenseLevel.Medium)]  
[MarkerAttributes.Complexity(MarkerAttributes.Complexity.TimeComplexity.ON)]  
public static void RemoveScene(string __name)
```

Parameters

__name [string](#)[↗](#)

The name of the scene you want to remove

Run()

Begins the update loop and sends Load() to all Components!

```
public static void Run()
```

StartCoroutine(IEnumerable<ICoroutineAction>)

Starts a coroutine

```
public static void StartCoroutine(IEnumerable<ICoroutineAction> __coroutine)
```

Parameters

__coroutine [IEnumerable](#)<[ICoroutineAction](#)>

VirtualClose()

Pretends to close Osmium, and makes the Components think that the Game has ended.

```
public static void VirtualClose()
```

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for the Editor! So don't use it unless you know what you are doing. If you do want to use it, use the EditorInitialize() EditorRun() and EditorClose() instead of the traditional methods!

VirtualInitialize(IEnumerable<Assembly>)

Pretends to initialize Osmium, and makes the Components think that the Game has just been initialized.

```
public static void VirtualInitialize(IEnumerable<Assembly> __assemblies)
```

Parameters

__assemblies [IEnumerable](#)<[Assembly](#)>

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for Radium! So don't use it unless you know what you are doing. If you do want to use it, use the EditorInitialize() EditorRun() and EditorClose() instead of the traditional methods!

VirtualRun()

Pretends to run Osmium virtually, and makes the Components think it has Started.

```
public static void VirtualRun()
```

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for Radium! So don't use it unless you know what you are doing. If you do want to use it, use the EditorInitialize() EditorRun() and EditorClose() instead of the traditional methods!

Events

Draw

```
public static event Action? Draw
```

Event Type

[Action](#)

FinalDraw

An event that is raised after all draw calls, this is meant to be used for libraries that require overhead

```
public static event Action? FinalDraw
```

Event Type

[Action](#)

FinalLoad

An event that is raised after all load calls, this is meant to be used for libraries that require overhead

```
public static event Action? FinalLoad
```

Event Type

[Action](#)

FinalUnload

An event that is raised after all unload calls, this is meant to be used for libraries that require overhead

```
public static event Action? FinalUnload
```

Event Type

[Action](#)

FinalUpdate

An event that is raised after all update calls, this is meant to be used for libraries that require overhead

```
public static event Action? FinalUpdate
```

Event Type

[Action](#)

FirstDraw

An event that is raised before all draw calls, this is meant to be used for libraries that require overhead

```
public static event Action? FirstDraw
```

Event Type

[Action](#)

FirstLoad

An event that is raised before all load calls, this is meant to be used for libraries that require overhead

```
public static event Action? FirstLoad
```

Event Type

[Action](#)

FirstUnload

An event that is raised before all unload calls, this is meant to be used for libraries that require overhead

```
public static event Action? FirstUnload
```

Event Type

[Action](#)

FirstUpdate

An event that is raised before all update calls, this is meant to be used for libraries that require overhead

```
public static event Action? FirstUpdate
```

Event Type

[Action](#)

Load

```
public static event Action? Load
```

Event Type

[Action](#)

SceneAdded

Is called when a new [Scene](#) is added!

```
public static event Action<Scene>? SceneAdded
```

Event Type

[Action](#) <[Scene](#)>

SceneRemoved

Is called when a new [Scene](#) is removed!

```
public static event Action<Scene>? SceneRemoved
```

Event Type

[Action](#) <[Scene](#)>

Unload

```
public static event Action? Unload
```

Event Type

[Action](#)

Update

```
public static event Action? Update
```

Event Type

[Action](#)

Class Scene

Namespace: [OsmiumNucleus](#)

Assembly: Osmium-Nucleus.dll

One of the main Osmium entities. Acts like a master folder for components to be stored in. Feel free to inherit this and make a custom scene!

```
public class Scene : ComponentDocker, IEnumerable<Component>, IEnumerable
```

Inheritance

[object](#)  ← [ComponentDocker](#) ← Scene

Implements

[IEnumerable](#)  <[Component](#)>, [IEnumerable](#) 

Inherited Members

[ComponentDocker.components](#), [ComponentDocker.componentPriorityDictionary](#),
[ComponentDocker.componentTypeDictionary](#), [ComponentDocker.componentTagDictionary](#),
[ComponentDocker.this\[int\]](#), [ComponentDocker.GetEnumerator\(\)](#),
[ComponentDocker.ComponentMoved](#), [ComponentDocker.ComponentAdded](#),
[ComponentDocker.ComponentDestroyed](#),
[ComponentDocker.Contains\(params IEnumerable<Component>\)](#),
[ComponentDocker.Contains<TComponent>\(\)](#),
[ComponentDocker.Contains\(params IEnumerable<string>\)](#),
[ComponentDocker.Contains<TComponent>\(params IEnumerable<string>\)](#), [ComponentDocker.Get\(\)](#),
[ComponentDocker.Get<TComponent>\(\)](#), [ComponentDocker.Get\(params IEnumerable<string>\)](#),
[ComponentDocker.Get<TComponent>\(params IEnumerable<string>\)](#), [ComponentDocker.GetAll\(\)](#),
[ComponentDocker.GetAll<TComponent>\(\)](#), [ComponentDocker.GetAll\(params IEnumerable<string>\)](#),
[ComponentDocker.GetAll<TComponent>\(IEnumerable<string>\)](#),
[ComponentDocker.Add\(params IEnumerable<Component>\)](#),
[ComponentDocker.Add<TComponent>\(\)](#), [ComponentDocker.Add<TComponent>\(string\)](#),
[ComponentDocker.Add<TComponent>\(int\)](#), [ComponentDocker.Add<TComponent>\(string,int\)](#),
[ComponentDocker.Destroy\(params IEnumerable<Component>\)](#),
[ComponentDocker.Destroy<TComponent>\(\)](#),
[ComponentDocker.Destroy\(params IEnumerable<string>\)](#),
[ComponentDocker.Destroy<TComponent>\(params IEnumerable<string>\)](#),
[ComponentDocker.DestroyAll\(\)](#), [ComponentDocker.DestroyAll<TComponent>\(\)](#),
[ComponentDocker.DestroyAll\(params IEnumerable<string>\)](#),

[ComponentDocker.DestroyAll<TComponent>\(params IEnumerable<string>\)](#) ,
[ComponentDocker.Move\(ComponentDocker, params IEnumerable<Component>\)](#) ,
[ComponentDocker.Move<TComponent>\(ComponentDocker\)](#) ,
[ComponentDocker.Move\(ComponentDocker, params IEnumerable<string>\)](#) ,
[ComponentDocker.Move<TComponent>\(ComponentDocker, params IEnumerable<string>\)](#) ,
[ComponentDocker.MoveAll\(ComponentDocker\)](#) ,
[ComponentDocker.MoveAll<TComponent>\(ComponentDocker\)](#) ,
[ComponentDocker.MoveAll\(ComponentDocker, params IEnumerable<string>\)](#) ,
[ComponentDocker.MoveAll<TComponent>\(ComponentDocker, params IEnumerable<string>\)](#) ,
[object.Equals\(object\)](#)[↗] , [object.Equals\(object, object\)](#)[↗] , [object.GetHashCode\(\)](#)[↗] ,
[object.GetType\(\)](#)[↗] , [object.MemberwiseClone\(\)](#)[↗] , [object.ReferenceEquals\(object, object\)](#)[↗] ,
[object.ToString\(\)](#)[↗]

Constructors

Scene(string)

One of the main Osmium entities. Acts like a master folder for components to be stored in. Feel free to inherit this and make a custom scene!

```
public Scene(string __name)
```

Parameters

__name [string](#)[↗]

Fields

Enabled

Whether the Scene is capturing events or not

```
public bool Enabled
```

Field Value

[bool](#)[↗]

Properties

Name

Unique identifier of the Scene

```
public string Name { get; set; }
```

Property Value

[string](#) 

Namespace OsmiumTests

Classes

[DockerTests](#)

[Tests](#)

Class DockerTests

Namespace: [OsmiumTests](#)

Assembly: Osmium-Tests.dll

```
[TestFixture]
public class DockerTests
```

Inheritance

[object](#)  ← DockerTests

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Methods

ComponentsMoveBetweenScenes()

```
[Test]
public void ComponentsMoveBetweenScenes()
```

FilledDockerContainsComponent()

```
[Test]
public void FilledDockerContainsComponent()
```

GetReturnsIdenticalReference()

```
[Test]
public void GetReturnsIdenticalReference()
```

GetTagReturnsNull()

```
[Test]  
public void GetTagReturnsNull()
```

IndexingMatchesChildren()

```
[Test]  
public void IndexingMatchesChildren()
```

Class Tests

Namespace: [OsmiumTests](#)

Assembly: Osmium-Tests.dll

```
public class Tests
```

Inheritance

[object](#)  ← Tests

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 